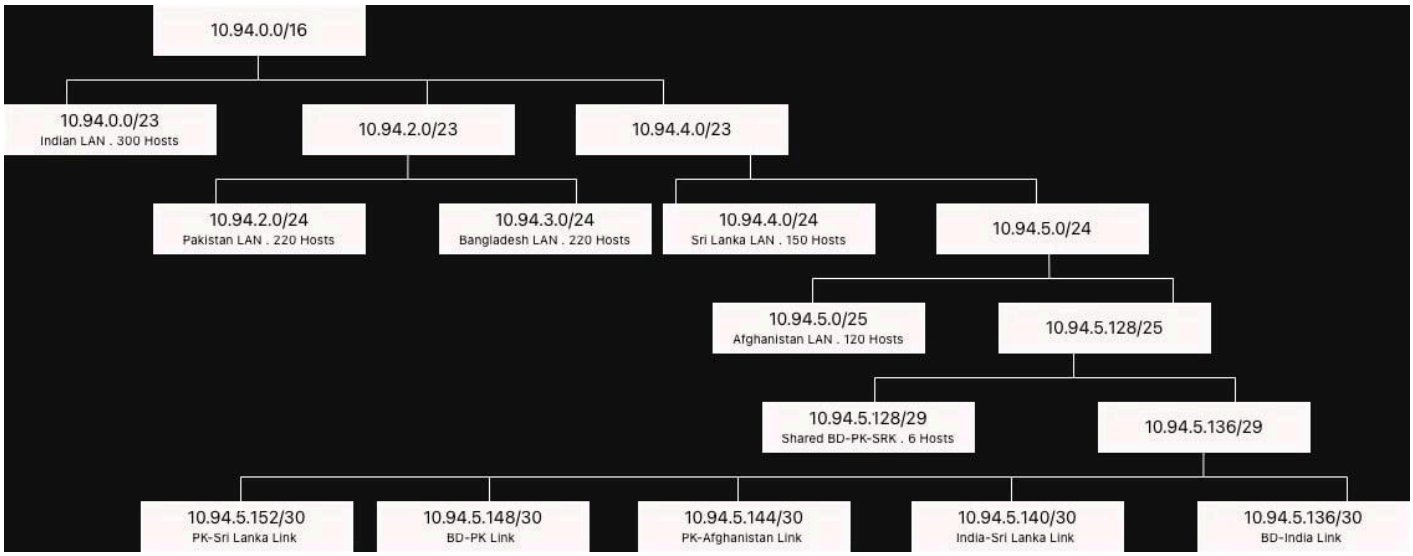
CRICKET WORLD CUP 2027

VLSM TREE, IP ADDRESS TABLE, ROUTER CONFIGURATION COMMANDS

VLSM TREE :



IP Address Table:

Network/Segment	Hosts Needed	Subnet	Subnet Mask	Gateway/Router IP(s)	DHCP/Static Strategy
India LAN	300	10.94.0.0/23	255.255.254.0	Router_IND Fa0/0 = 10.94.0.1	India DHCP server; PC range starts 10.94.0.10
Pakistan LAN	220	10.94.2.0/24	255.255.255.0	Router_PK Fa0/0 = 10.94.2.1	Bangladesh router DHCP via helper; PC range 10.94.2.10-10.94.2.229
Bangladesh LAN	180	10.94.3.0/24	255.255.255.0	Router_BD Fa0/0 = 10.94.3.1	Bangladesh router DHCP; PC range 10.94.3.10-10.94.3.189
Sri Lanka LAN	150	10.94.4.0/24	255.255.255.0	Router_SRK Fa0/0 = 10.94.4.1	Bangladesh router DHCP via helper; PC range 10.94.4.10-10.94.4.159
Afghanistan LAN	120	10.94.5.0/25	255.255.255.128	Router_AFG Fa0/0 = 10.94.5.1	Afghanistan DHCP server; PC range 10.94.5.2-10.94.5.123
BD-PK-SRK shared switch	3 routers	10.94.5.128/29	255.255.255.248	BD .129, PK .130, SRK .131	Router interfaces only; switch is Layer 2
BD-IND P2P	2	10.94.5.136/30	255.255.255.252	BD .137, IND .138	Serial link
IND-SRK P2P	2	10.94.5.140/30	255.255.255.252	IND .141, SRK .142	Serial link
PK-AFG P2P	2	10.94.5.144/30	255.255.255.252	PK .145, AFG .146	Serial link
BD-PK Direct P2P	2	10.94.5.148/30	255.255.255.252	BD .149, PK .150	Serial link
PK-SRK Direct P2P	2	10.94.5.152/30	255.255.255.252	PK .153, SRK .154	Serial link

Router Interface IP Table :

Router	Interface	Connected To	IP Address	Subnet Mask
Router_BD	FastEthernet0/0	Bangladesh LAN	10.94.3.1	255.255.255.0
Router_BD	FastEthernet0/1	Shared BD-PK-SRK switch	10.94.5.129	255.255.255.248
Router_BD	Serial0/0/0	India	10.94.5.137	255.255.255.252
Router_BD	Serial0/0/1	Pakistan direct	10.94.5.149	255.255.255.252
Router_IND	FastEthernet0/0	India LAN	10.94.0.1	255.255.254.0
Router_IND	Serial0/0/0	Bangladesh	10.94.5.138	255.255.255.252
Router_IND	Serial0/0/1	Sri Lanka	10.94.5.141	255.255.255.252
Router_PK	FastEthernet0/0	Pakistan LAN	10.94.2.1	255.255.255.0
Router_PK	FastEthernet0/1	Shared BD-PK-SRK switch	10.94.5.130	255.255.255.248
Router_PK	Serial0/0/0	Afghanistan	10.94.5.145	255.255.255.252
Router_PK	Serial0/1/0	Bangladesh direct	10.94.5.150	255.255.255.252
Router_PK	Serial0/0/1	Sri Lanka direct	10.94.5.153	255.255.255.252
Router_SRK	FastEthernet0/0	Sri Lanka LAN	10.94.4.1	255.255.255.0
Router_SRK	FastEthernet0/1	Shared BD-PK-SRK switch	10.94.5.131	255.255.255.248
Router_SRK	Serial0/0/0	India	10.94.5.142	255.255.255.252
Router_SRK	Serial0/0/1	Pakistan direct	10.94.5.154	255.255.255.252
Router_AFG	FastEthernet0/0	Afghanistan LAN	10.94.5.1	255.255.255.128
Router_AFG	Serial0/0/0	Pakistan	10.94.5.146	255.255.255.252

Static Server and Printer IP Table :

Country	Device	IP Address	Subnet Mask	Default Gateway	DNS Server
Bangladesh	Web Server	10.94.3.254	255.255.255.0	10.94.3.1	10.94.2.254
Bangladesh	Email Server	10.94.3.253	255.255.255.0	10.94.3.1	10.94.2.254
Bangladesh	Printer	10.94.3.252	255.255.255.0	10.94.3.1	10.94.2.254
India	Web Server	10.94.1.254	255.255.254.0	10.94.0.1	10.94.2.254
India	Email Server	10.94.1.253	255.255.254.0	10.94.0.1	10.94.2.254
India	DHCP Server	10.94.1.252	255.255.254.0	10.94.0.1	10.94.2.254
India	Printer	10.94.1.251	255.255.254.0	10.94.0.1	10.94.2.254
Pakistan	DNS Server	10.94.2.254	255.255.255.0	10.94.2.1	10.94.2.254
Pakistan	Email Server	10.94.2.253	255.255.255.0	10.94.2.1	10.94.2.254
Pakistan	Printer	10.94.2.252	255.255.255.0	10.94.2.1	10.94.2.254
Sri Lanka	Email Server	10.94.4.254	255.255.255.0	10.94.4.1	10.94.2.254
Sri Lanka	Printer	10.94.4.253	255.255.255.0	10.94.4.1	10.94.2.254
Afghanistan	Email Server	10.94.5.126	255.255.255.128	10.94.5.1	10.94.2.254
Afghanistan	DHCP Server	10.94.5.125	255.255.255.128	10.94.5.1	10.94.2.254
Afghanistan	Printer	10.94.5.124	255.255.255.128	10.94.5.1	10.94.2.254

DHCP and DNS Table :

DHCP Source	Served LAN	Default Gateway	DNS Server	DHCP Client Range / Settings
Router_BD	Bangladesh LAN	10.94.3.1	10.94.2.254	10.94.3.10 - 10.94.3.189
Router_BD via Router_PK helper	Pakistan LAN	10.94.2.1	10.94.2.254	10.94.2.10 - 10.94.2.229
Router_BD via Router_SRK helper	Sri Lanka LAN	10.94.4.1	10.94.2.254	10.94.4.10 - 10.94.4.159
India DHCP Server	India LAN	10.94.0.1	10.94.2.254	Start 10.94.0.10, max users 300
Afghanistan DHCP Server	Afghanistan LAN	10.94.5.1	10.94.2.254	Start 10.94.5.2, max users 122

Pakistan Central DNS Server and DNS Records :

Server Device Name	Purpose	IP Address	Default Gateway	DNS Setting
PK_dns_server	Central DNS Server	10.94.2.254	10.94.2.1	10.94.2.254

DNS Type	Domain/Host Name	IP Address	Points To
A Record	www.bangladesh.cricket.com	10.94.3.254	Bangladesh Web Server
A Record	mail.bangladesh.com	10.94.3.253	Bangladesh Email Server
A Record	www.india.cricket.com	10.94.1.254	India Web Server
A Record	mail.india.com	10.94.1.253	India Email Server
A Record	mail.pakistan.com	10.94.2.253	Pakistan Email Server
A Record	mail.srilanka.com	10.94.4.254	Sri Lanka Email Server
A Record	mail.afghanistan.com	10.94.5.126	Afghanistan Email Server

ROUTER CONFIGURATION COMMANDS :

Router_BD (Bangladesh Router):

```
enable
configure terminal
hostname Router_BD
interface fastEthernet 0/0
no ip address
ip address 10.94.3.1 255.255.255.0
no shutdown
exit
interface fastEthernet 0/1
no ip address
ip address 10.94.5.129 255.255.255.248
```

```
no shutdown
exit
interface serial 0/0/0
no ip address
ip address 10.94.5.137 255.255.255.252
no shutdown
exit
interface serial 0/0/1
no ip address
ip address 10.94.5.149 255.255.255.252
no shutdown
exit
service dhcp
ip dhcp excluded-address 10.94.3.1 10.94.3.9
ip dhcp excluded-address 10.94.3.190 10.94.3.254
ip dhcp excluded-address 10.94.2.1 10.94.2.9
ip dhcp excluded-address 10.94.2.230 10.94.2.254
ip dhcp excluded-address 10.94.4.1 10.94.4.9
ip dhcp excluded-address 10.94.4.160 10.94.4.254
ip dhcp pool BANGLADESH_LAN
network 10.94.3.0 255.255.255.0
default-router 10.94.3.1
dns-server 10.94.2.254
exit
ip dhcp pool PAKISTAN_LAN
network 10.94.2.0 255.255.255.0
default-router 10.94.2.1
```

dns-server 10.94.2.254

exit

ip dhcp pool SRILANKA_LAN

network 10.94.4.0 255.255.255.0

default-router 10.94.4.1

dns-server 10.94.2.254

exit

no router rip

router rip

version 2

no auto-summary

network 10.0.0.0

passive-interface fastEthernet 0/0

passive-interface serial 0/0/1

exit

ip route 10.94.2.0 255.255.255.0 10.94.5.130

ip route 10.94.4.0 255.255.255.0 10.94.5.131

ip route 10.94.5.0 255.255.255.128 10.94.5.130

ip route 10.94.5.144 255.255.255.252 10.94.5.130

ip route 10.94.5.152 255.255.255.252 10.94.5.130

end

copy running-config startup-config

Router_IND (India Router) :

enable

configure terminal

```
hostname Router_IND
interface fastEthernet 0/0
no ip address
ip address 10.94.0.1 255.255.254.0
no shutdown
exit
interface serial 0/0/0
no ip address
ip address 10.94.5.138 255.255.255.252
no shutdown
exit
interface serial 0/0/1
no ip address
ip address 10.94.5.141 255.255.255.252
no shutdown
exit
no router rip
router rip
version 2
no auto-summary
network 10.0.0.0
passive-interface fastEthernet 0/0
exit
ip route 10.94.3.0 255.255.255.0 10.94.5.137
ip route 10.94.3.0 255.255.255.0 10.94.5.142 5
ip route 10.94.2.0 255.255.255.0 10.94.5.142
ip route 10.94.5.0 255.255.255.128 10.94.5.142
```

ip route 10.94.5.144 255.255.255.252 10.94.5.142

ip route 10.94.5.152 255.255.255.252 10.94.5.142

end

copy running-config startup-config

Router_PK (Pakistan Router) :

enable

configure terminal

hostname Router_PK

interface fastEthernet 0/0

no ip address

ip address 10.94.2.1 255.255.255.0

ip helper-address 10.94.5.129

no shutdown

exit

interface fastEthernet 0/1

no ip address

ip address 10.94.5.130 255.255.255.248

no shutdown

exit

interface serial 0/0/0

no ip address

ip address 10.94.5.145 255.255.255.252

no shutdown

exit

interface serial 0/1/0

no ip address

ip address 10.94.5.150 255.255.255.252

no shutdown

exit

interface serial 0/0/1

no ip address

ip address 10.94.5.153 255.255.255.252

no shutdown

exit

no router rip

ip route 10.94.0.0 255.255.254.0 10.94.5.154

ip route 10.94.3.0 255.255.255.0 10.94.5.149

ip route 10.94.4.0 255.255.255.0 10.94.5.154

ip route 10.94.5.0 255.255.255.128 10.94.5.146

ip route 10.94.5.136 255.255.255.252 10.94.5.149

ip route 10.94.5.140 255.255.255.252 10.94.5.154

end

copy running-config startup-config

Router_SRK (Sri Lanka Router) :

enable

configure terminal

hostname Router_SRK

interface fastEthernet 0/0

no ip address

ip address 10.94.4.1 255.255.255.0

```
ip helper-address 10.94.5.129
no shutdown
exit
interface fastEthernet 0/1
no ip address
ip address 10.94.5.131 255.255.255.248
no shutdown
exit
interface serial 0/0/0
no ip address
ip address 10.94.5.142 255.255.255.252
no shutdown
exit
interface serial 0/0/1
no ip address
ip address 10.94.5.154 255.255.255.252
no shutdown
exit
no router rip
router rip
version 2
no auto-summary
network 10.0.0.0
passive-interface fastEthernet 0/0
passive-interface serial 0/0/1
exit
ip route 10.94.2.0 255.255.255.0 10.94.5.153
```

ip route 10.94.5.0 255.255.255.128 10.94.5.153

ip route 10.94.5.144 255.255.255.252 10.94.5.153

ip route 10.94.5.148 255.255.255.252 10.94.5.153

end

copy running-config startup-config

Router_AFG (Afghanistan Router) :

enable

configure terminal

hostname Router_AFG

interface fastEthernet 0/0

no ip address

ip address 10.94.5.1 255.255.255.128

no shutdown

exit

interface serial 0/0/0

no ip address

ip address 10.94.5.146 255.255.255.252

no shutdown

exit

no router rip

```
ip route 0.0.0.0 0.0.0.0 10.94.5.145
end
copy running-config startup-config
```